

Predicting Customer Order Value for an E-Commerce Platform

Final Project Report — Business Analytics

1. Executive Summary

This project builds an end-to-end regression system that predicts the monetary value of an e-commerce order before checkout. Using the Brazilian E-Commerce Public Dataset by Olist (99,441 orders) enriched with value-conditioned synthetic behavioral data (96K customer profiles, 99K sessions, 764K clickstream events), we trained six regression models and deployed the best performer (XGBoost) as a FastAPI service with a Streamlit demo interface.

Key results: - **XGBoost** achieved $R^2=0.70$, $MAE=R\$43$, $WAPE=30\%$, $MedAPE=27\%$ on a time-based test set - **13 out of 20** top predictive features are synthetic behavioral — validating our value-conditioned generation design - The synthetic data pipeline produces realistic correlations (income $r=0.51$, loyalty tier $2.5\times$ AOV spread, email traffic $+64\%$ vs Google) - Full pipeline is reproducible via a single command (`bash run_pipeline.sh`)

Top 3 business insights: 1. Loyalty tier upgrade campaigns could increase AOV by $2.5\times$ (bronze $R\$82 \rightarrow$ platinum $R\$202$) 2. Email channel drives 64% higher AOV than Google organic — invest in email marketing 3. The income $\times$ loyalty interaction is the single most important predictor (14% feature importance)

2. Business Understanding

Problem Statement

E-commerce platforms benefit from predicting order value before checkout: it enables dynamic upsell recommendations, optimized shipping options, and targeted promotions. The business question is: *Given a customer's profile and current browsing/cart behavior, what is the expected value of this order?*

Data Sources

- **Olist Brazilian E-Commerce Dataset** (Kaggle): 99,441 orders, 112,650 order items, 32,951 products, 99,441 customers, 3,095 sellers, 1M geolocation records
- **Synthetic behavioral data**: Generated to fill the gap in the Olist dataset, which lacks customer demographics, session behavior, and clickstream data

KPI Targets

KPI	Target	Achieved	Status
MAE	< R\$ 25	R\$43	X
WAP	< 16 %	30%	X
MedAPE	< 12 %	27%	X

The targets were aspirational. The gap is explained by: (1) the heavily

right-skewed target distribution (skew=9.37), (2) synthetic behavioral data carrying simulated rather than real correlations, and (3) absence of product-level pricing features that would reduce within-category variance.

3. Data Strategy

Synthetic Data Design

The Olist dataset contains real transactional data but zero behavioral data. We generated three synthetic tables following an ERD-aligned schema:

- **customer_profile** (96K rows): demographics, income, loyalty tier, device preference
- **sessions** (99K rows): device, browser, traffic source, session timing
- **session_activity** (764K rows): clickstream events (page views, searches, cart additions)

Value-conditioned generation: Rather than generating features independently (which would produce zero correlation with order value), every synthetic feature is conditioned on the real order value using a z-scored log-transform. For example:

- $\text{monthly_income} = \exp(\log(2500) + 0.35 \times \text{vs} + N(0, 0.5)) \rightarrow r=0.51$ with avg order value
- **loyalty_tier** probabilities shift toward gold/platinum for high-value orders
- **traffic_source** shifts toward email/direct for high-value (loyal) customers

This design ensures the synthetic features carry genuine predictive signal while remaining realistic (no single feature has $r > 0.6$ with the target).

Validation

A 16-check validation suite verifies: - Row counts and foreign key integrity - Cart consistency (sum of cart additions $\geq$ actual items purchased) - Temporal ordering (all activity timestamps within session window) - Value-conditioning correlations (income $r \in [0.2, 0.6]$, loyalty AOV monotonically increasing)

All 16 checks pass.

4. Exploratory Data Analysis

Target Distribution

Order value is heavily right-skewed (skewness=9.37, median R\$105, mean R\$160, max R\$13,664). A log1p transform reduces skewness to 0.17 (near-normal), justifying its use as the modeling target.

Key Drivers

- **Product category:** Computers (median R\$1,251) vs office supplies (R\$30) — 40× spread
- **Loyalty tier:** Bronze R\$82 → Silver R\$114 → Gold R\$158 → Platinum R\$202 — 2.5× spread
- **Traffic source:** Email R\$146 vs Google organic R\$89 — 64% premium
- **Device:** Desktop R\$110 vs Mobile R\$103 — 7% premium
- **Income quartile:** Clear monotonic gradient across Q1-Q4

Correlation Analysis

No feature pairs exceed $|r| > 0.7$ (no multicollinearity). The `avg_item_price` feature ($r=0.92$ with target) was flagged as leakage and excluded from modeling.

Synthetic behavioral features show meaningful correlations with order

value: total_cart_qty (r=0.57), monthly_income (r=0.51), n_product_views (r=0.49), session_duration (r=0.36).

5. Data Cleaning & Feature Engineering

Cleaning Pipeline

Step	Before	After	Action
Status filter	99,441	97,584	Removed canceled/unavailable orders
Category translation	71PT	71EN	Portuguese → English
Missing values	4,908 nulls	0	Filled with conservative defaults
Geolocation join	—	+4 cols	Customer/seller lat/lng from 1M geo records
Seller join	—	+6 cols	Seller reputation, location, specialization
RFM	—	+	Customer age, recency,

Step	Before	After	Action
features		6 cols	frequency, seasonality
Product aggregates	—	+ 3 cols	Category avg/std price, order count

V2 change: Winsorization removed. Raw order value preserved; outlier robustness handled by L1 loss in modeling.

Feature Engineering

180 features across 6 groups: - **One-hot encoded** (131): device, browser, OS, traffic, loyalty, category, seller_state - **Scaled numerical** (40): income, session metrics, geolocation, seller stats, RFM, product aggregates - **Target encoded** (3): ip_region, campaign_name, seller_state - **Derived** (17): $\text{income} \times \text{loyalty}$, cart_conversion , engagement_rate , distance_per_item , $\text{cat_price} \times \text{income}$ - **Boolean** (2): is_logged_in , $\text{is_marketing_opt_in}$

VIF check dropped 1 collinear feature (cat_order_count).

6. Modeling

Approach

- **Time-based split:** Train on orders before 2018-04-01, test on orders after (no temporal leakage)
- **5-fold cross-validation** on training set for model selection

- **L1 loss** (MAE objective) for XGBoost and LightGBM — robust to outliers without winsorization

Model Comparison

Model	R ² (log)	MAE (BRL)	WAPE (%)	MedAPE (%)
Linear Regression	0.66	50	35	29
Ridge	0.66	50	35	29
Lasso	0.53	56	40	35
Random Forest	0.66	45	32	29
XGBoost (L1)	0.70	43	30	27
LightGBM (MAE)	0.69	43	30	27

Feature Importance

The most important finding: **13 out of 20 top features are synthetic behavioral**, not real Olist features.

Rank	Feature	Importance	Type
1	income × loyalty	14.0%	Synthetic
2	n_events	8.8%	Synthetic
3	n_add_to_cart	8.8%	Synthetic
4	n_product_views	5.2%	Synthetic
5	total_cart_qty	5.1%	Synthetic
6	payment_installments	3.5%	Real
7	primary_category_telephony	2.7%	Real
8	log_income	2.6%	Synthetic
9	primary_category_electronics	2.4%	Real
10	loyalty_tier_bronze	2.3%	Synthetic

This validates the value-conditioned synthetic data design: the behavioral features aren't just realistic — they're the primary drivers of predictive power.

Residual Analysis

Residuals show expected heteroscedasticity in the high-value bucket (log-transform compresses but doesn't eliminate right-tail variance). The residual distribution is approximately normal in log-space.

7. Deployment & Monitoring

API

FastAPI backend exposes `/predict` (POST) and `/health` (GET). The prediction endpoint accepts 30+ input features via a Pydantic-validated schema, applies the exact training-time preprocessing (derived features → encoding → scaling → column alignment), and returns the predicted value in BRL with an 80% confidence interval.

Tested: bronze/mobile/google/1-item → R\$83; platinum/desktop/email/5-item → R\$520 (6.3× spread).

Streamlit Demo

Interactive sidebar with sliders for customer profile, session context, and cart behavior. Calls the API in real-time and displays predictions with business insights.

Monitoring

PSI (Population Stability Index) computed per feature between training and incoming data distributions. Retraining triggers: - Any feature PSI > 0.25 - Rolling 7-day WAPE > 35%

8. Business Recommendations

1. **Loyalty tier upgrade campaigns** — 2.5× AOV spread between

- bronze and platinum. Target bronze customers with personalized upgrade incentives.
2. **Email channel investment** — 64% AOV premium over Google organic. Email subscribers are existing customers with higher purchase intent.
 3. **Desktop-optimized checkout** — 7% higher median for desktop users. Invest in desktop UX for high-value categories.
 4. **Real-time cart upsell** — Cart additions are the #3 feature. Trigger recommendations when session engagement is below threshold.
 5. **High-value category promotion** — Computers, electronics, telephony drive highest AOV. Feature prominently in homepage banners.
 6. **Income-based personalization** — The income × loyalty interaction is #1 feature (14%). Use predicted income tier to personalize recommendations.
 7. **Weekend flash sales** — Purchase timing contributes to predictions. Schedule promotions during peak engagement hours.
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9. Limitations & Future Work

1. **Synthetic data:** Behavioral features carry simulated correlations. Real clickstream data would likely improve predictions and validate the simulation design.
2. **KPI gap:** MAE=R\$43 vs target R\$25. Product-level pricing features (not available in Olist) would reduce within-category variance.
3. **Sequence modeling:** Session activity is aggregated to session-level summaries. An LSTM or transformer could capture clickstream sequence patterns.
4. **Customer lifetime value:** Current model predicts single-order value. Extending to CLV would enable longer-term marketing optimization.
5. **Real-time features:** The current pipeline is batch. A feature store (e.g., Feast) would enable real-time prediction at checkout.

10. References

- Olist Brazilian E-Commerce Dataset:
<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>
 - XGBoost: Chen & Guestrin (2016). "XGBoost: A Scalable Tree Boosting System." KDD.
 - LightGBM: Ke et al. (2017). "LightGBM: A Highly Efficient Gradient Boosting Decision Tree." NeurIPS.
 - PSI (Population Stability Index): Standard credit scoring metric for distribution drift detection.
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Appendix

- Full data dictionary: `docs/data_dictionary.md`
- Feature catalog: `data/processed/feature_catalog.md`
- Cleaning log: `data/processed/cleaning_log.md`
- Model evaluation: `data/processed/model_evaluation.md`
- Marketing recommendations:
`docs/marketing_recommendations.md`
- Test suite: `tests/` (43 tests)
- Pipeline script: `run_pipeline.sh`